
BUSA 451: Faculty Research Project

Lecture Notes & Research Roadmap

Quantifying the Value of Arterial Stiffness Information for Optimal Hypertension Management

Integrated Management Student Fellowship (IMSF) — 2026–2027 Cohort
Marcel Desautels Institute for Integrated Management (MDIIM)

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These notes translate the Faculty Research Project component of the Integrated Management Student Fellowship into a module-by-module work plan for the 2026–2027 academic year, prepared in accordance with the IMSF Faculty Supervisors Onboarding Document and the project introduction confirming the research match.

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This document is the working syllabus for the Faculty Research Project component of BUSA 451 (D1, D2), prepared for IMSF FellowS under the co-supervision of Professor Mehmet Gumus (Operations Management Area, Desautels Faculty of Management) and Professor Stella Daskalopoulou (Division of Internal Medicine, Department of Medicine). The underlying project, *Quantifying the Value of Arterial Stiffness Information for Optimal Hypertension Management*, is a McGill–MUHC research program that studies whether a second vascular measurement (arterial stiffness, or cfPWV) should change how doctors treat high blood pressure. That research program is led by the Supervisors and is technically advanced. Fellows contribute to the parts of the research process that are foundational to *any* data-driven project: collecting and documenting data, cleaning and organizing it, applying existing prediction tools correctly, and using AI tools responsibly to support the team’s work – the same data-foundation activities that any serious data-science or health-analytics initiative must get right before more advanced modelling can proceed.

Day-to-day administration of the Fellows’ work is handled by Kibele Sebnem Yildirim, a PhD student supervised by Professor Gumus, who serves as the project’s Technical Project Lead under his supervision. Kibele coordinates the modules described below, reviews Fellow deliverables in the first instance, and is each Fellow’s primary point of contact for technical questions day to day; Professors Gumus and Daskalopoulou remain the Faculty Supervisors of record and set overall project direction. See “Project Team & Roles” below for the full breakdown of responsibilities and meeting cadence.

The Fellowship runs two semesters (September 2026 – April 2027, six credits) and is organized here into eight modules, four per semester, aligned with the McGill academic calendar. Each module states its learning objectives, concrete and approachable tasks, the division of responsibility between Fellows and Supervisors, the deliverable(s) due, and the estimated hours, consistent with the 60 hours per semester (120 hours per academic year) that each Fellow is expected to contribute.

Project Team & Roles

Role	Name	Meets with Fellows
Faculty Supervisor (project direction)	Professor Mehmet Gumus, Operations Management Area	Bi-weekly (or as needed)
Faculty Co-Supervisor (clinical direction)	Professor Stella Daskalopoulou, Division of Internal Medicine	Bi-weekly (or as needed)
Technical Project Lead (day-to-day administration)	Kibele Sebnem Yildirim, PhD Student, supervised by Professor Gumus	Weekly

A note on the Technical Project Lead. Kibele Sebnem Yildirim is a PhD student supervised by Professor Gumus. She administers this project under his supervision and is the Fellows’ first point of contact for day-to-day technical questions, module

deliverables, and scheduling. Because the working relationship with the Technical Project Lead is the most frequent one Fellows will have, Fellows are expected to meet with her **weekly** – more frequently than the bi-weekly cadence Fellows keep with the Faculty Supervisors.

Logistics note. Each Fellow contributes 60 hours per semester (120 hours/year) to the faculty project. Fellows meet **weekly** with the Technical Project Lead (Sebnem Kibele Yildirim) and **bi-weekly** (or as needed) with the Faculty Supervisors to review progress, give feedback, and set next steps. Fellows must respect the Fellowship Research Agreement and keep a detailed logbook of contributions and hours throughout the year. Supervisors submit a performance evaluation of each Fellow to IMSF Program Leadership at the end of each semester.

Kickoff deadline. Per the IMSF onboarding schedule, Faculty Supervisors reach out to Fellows to schedule an initial meeting **by Friday, September 11, 2026**. Sebnem Kibele Yildirim, the Technical Project Lead, will be introduced at this meeting. The initial meeting must (i) set a recurring weekly meeting time with the Technical Project Lead and a bi-weekly (or as-needed) meeting time with the Faculty Supervisors, (ii) clarify how deliverables will be submitted and reviewed, and (iii) confirm the tools to be used (Python, shared OneDrive/GitHub repository, reference manager, etc.).

Roadmap at a Glance

Module	Focus	Window	Key Deliverable	Hrs
M1	Onboarding, Data Ethics & Orientation	Sept 2026	Signed Research Agreement; ethics training done; tools set up	12
M2	Data Collection, Data Dictionary & Literature Review	Sept–Oct 2026	Data dictionary + short literature review (6 seminal papers)	16
M3	Data Cleaning & Quality Checks	Oct–Nov 2026	Cleaned dataset + data-cleaning log	16
M4	Building an Analysis-Ready Dataset & Fall Checkpoint	Nov–Dec 2026	Tidy analysis-ready dataset; Fall progress presentation	16
M5	Applying Existing Prediction Models with Python	Jan 2027	Prediction-formula notebook + accuracy comparison	16
M6	Using AI Tools to Support Research	Feb 2027	Fact-checked AI-assisted literature summaries	14
M7	Data Visualization & Communicating Results	Mar 2027	Chart set + visual summary document	16

Module	Focus	Window	Key Deliverable	Hrs
M8	Final Presentation & Data Package Handoff	Apr 2027	Final Fellow presentation; documented data package	14
Total (per Fellow, per year)				120

1 | Module 1: Onboarding, Data Ethics & Orientation to the Project

Overview

Module 1 gets IMSF Fellow(s) oriented to the project without requiring any prior clinical or advanced statistical background. The project studies hypertension (high blood pressure) and a second measurement called arterial stiffness (cfPWV), and asks whether knowing a patient's arterial stiffness changes what treatment a doctor should recommend. Fellows do not need to understand the full mathematical model behind this question. Their role over the year is to help the research team collect, clean, organize, and apply existing prediction tools to real patient data – the essential groundwork that any data-driven research project depends on. This module covers basic orientation, research ethics and data privacy, and getting comfortable with the tools (Python, spreadsheets, and shared folders) the team will use all year.

Learning Objectives

- Explain, in plain language, what blood pressure and arterial stiffness (cfPWV) are, and why researchers might want to measure both.
- Explain why patient data must be de-identified and handled carefully, and describe the basic rules of research ethics and confidentiality that apply to this project.
- Open, view, and navigate a spreadsheet or CSV file of patient data using Excel and Python (pandas).
- Describe, at a high level, what the research team means by “data collection,” “data cleaning,” and “using an AI/prediction model” – the three activities Fellows will practice this year.

Tasks

- Attend the initial kickoff meeting with Professors Gumus and Daskalopoulou (by **September 11, 2026** per the IMSF onboarding schedule).
- Read a short, non-technical two-page project summary (to be provided at kickoff) explaining blood pressure, arterial stiffness, and the research question in plain language.
- Complete McGill's basic research-ethics and data-privacy orientation (e.g., handling of de-identified health data, confidentiality expectations) before touching any patient

data.

- Install and test Python (via a shared cloud notebook environment such as Google Colab or JupyterHub) and the pandas library; complete a short guided exercise opening and viewing a sample (fake/practice) dataset.
- Set up the shared project folder (OneDrive/GitHub), the Fellowship logbook, and confirm meeting cadence with supervisors.
- Sign the Fellowship Research Agreement.

No prior coding or clinical background required. This module assumes no experience with Python, statistics, or medicine. Everything is taught from the ground up. What matters is care, curiosity, and attention to detail – the same qualities that make for good data work in any field.

Responsibilities

Fellows	Supervisors
Complete the ethics/privacy orientation and the Python starter exercise.	Provide the plain-language project summary and the ethics/privacy orientation materials.
Set up tools and shared workspace as directed.	Provide a practice dataset and a short guided walkthrough of Python/pandas basics.
Sign the Fellowship Research Agreement and begin the logbook.	Schedule the recurring bi-weekly meeting and confirm how work will be reviewed.

Deliverables & Deadline

Due by end of September 2026:

- Completed research-ethics and data-privacy orientation.
- Signed Fellowship Research Agreement (both Fellows).
- Working Python/pandas environment, tested on a practice dataset.
- Kickoff meeting held by **September 11, 2026**; recurring meeting cadence set.

Estimated Hours

12 hours per Fellow (orientation and ethics training: 4h; tooling setup and practice exercises: 5h; kickoff and follow-up meetings: 3h).

2 | Module 2: Data Collection, the Data Dictionary & a First Literature Review

Overview

Before any analysis can happen, a research team needs to know exactly what data it has, where it came from, and what each column means – and it needs to understand the handful of prior papers that motivate and justify the project in the first place. Module 2 introduces Fellows to two foundational research skills at once: *data collection*, meaning identifying and gathering the pieces of information that describe a patient’s visit and documenting them in a *data dictionary*; and a *first literature review*, meaning reading a small set of seminal papers and summarizing, in the Fellows’ own words, what each one found and why it matters to this project. Both skills are standard practice at the start of any data-driven research project, and both are squarely within undergraduate reach when scoped carefully, as they are here.

Learning Objectives

- Explain what a data dictionary is and why research teams build one before doing any analysis.
- Identify the categories of information typically collected about a patient visit: demographics (age, sex), measurements (blood pressure, arterial stiffness), medications, and visit/administrative details.
- Distinguish between raw data (as originally recorded) and processed data (cleaned and organized for analysis).
- Read an academic paper strategically (abstract, introduction, and conclusion first) and summarize its research question, its main finding, and why it matters, without needing to follow every mathematical or statistical detail.
- Explain, in plain language, why this project’s supervisors think arterial stiffness is worth studying alongside blood pressure, using the literature as evidence.

Tasks: Data Collection

- Meet with Prof. Daskalopoulou’s team to understand where the project’s patient data comes from and how it was originally recorded.

- Build a data dictionary for the project’s dataset: for every column, record its name, a plain-language description, its unit of measurement (e.g., mmHg, m/s, years), its type (number, category, date), and any known limitations (e.g., some patients are missing this measurement).
- Organize the data dictionary in a simple, shared spreadsheet that the whole team can reference throughout the year.
- Using the table below as a guide, sort the project’s variables into categories, mirroring how research teams typically organize patient data.

Example: How Patient Data Is Organized

The table below is a simplified example (not the real patient data) showing how research teams typically group the pieces of information collected about a patient visit. Fellows will build the actual version of this table for the project’s dataset as part of this module.

Demographics	Measurements	Visit / Treatment
Age	Systolic blood pressure (SBP)	Visit date
Sex	Diastolic blood pressure (DBP)	Medication level
Smoking status	Arterial stiffness (cfPWV)	Follow-up interval
Diabetes status	Total cholesterol, HDL	Clinic site

Tasks: Literature Review

Every research project stands on the shoulders of earlier work. Before building anything, Fellows will read six short, carefully chosen papers that explain *why* this project exists: what arterial stiffness is, why it might matter beyond blood pressure, how it is actually measured in a clinic, and the closest prior study (co-authored by Prof. Daskalopoulou) that first modeled hypertension treatment as a sequential decision problem. Fellows are not expected to follow every equation – the goal is to understand each paper’s motivating question, its main finding, and why the finding is relevant to this project.

- Read the six papers listed in Table 2.1 below. For each one, focus on the abstract, introduction, and conclusion/discussion sections first; skim (rather than work through in detail) any heavily mathematical or statistical sections.
- For each paper, write a short paragraph (4–6 sentences) answering: (i) what question was the paper trying to answer? (ii) what did it find? and (iii) why does this matter for a project studying arterial stiffness and hypertension treatment?
- Write one closing paragraph (the “synthesis”) connecting the six papers into a single narrative: what story do they tell, together, about why measuring arterial stiffness could change how patients are treated?
- Discuss the literature review with supervisors at a bi-weekly meeting, and revise based on their feedback.

Table 2.1: Literature review reading list for Module 2.

Paper	Why it is on the list
Zargoush et al. (2018)	The direct predecessor to this project, co-authored by Prof. Gumus and Prof. Daskalopoulou. It is the first paper to model hypertension treatment as a sequential decision problem and to show that a dynamically optimized treatment policy can outperform fixed guideline thresholds. Read for the motivation and big-picture findings, not the mathematical proofs.
Ben-Shlomo et al. (2014)	A large meta-analysis showing that arterial stiffness (cfPWV) predicts cardiovascular events (heart attack, stroke) even after accounting for blood pressure. This is the key clinical evidence that stiffness carries information blood pressure alone does not.
Vasan et al. (2022)	A long-term (15-year) Framingham Heart Study analysis confirming the same basic finding as above in a different, well-known patient cohort – a good example of how a result is strengthened when it replicates across studies.
Protogerou et al. (2009)	Shows that patients with stiffer arteries respond less well to blood-pressure-lowering medication. This motivates why the project treats stiffness as something that changes <i>how well treatment works</i> , not just as an extra risk factor.
Butlin and Qasem (2016)	A practical, technical-but-readable paper explaining how arterial stiffness is actually measured in a clinic (the SphygmoCor device used in this project’s patient cohort). Useful for understanding where the numbers in the dataset literally come from.
Howard (1966)	A short, classic paper that introduces the general idea of “value of information”: how much is it worth, in a decision problem, to learn something you currently do not know? This is the conceptual question behind the whole project, phrased in its simplest and most historically important form.

How to read a paper you are not yet trained for. Every one of these papers uses some specialized vocabulary or statistics. Read the abstract first to get the big picture, then the introduction for context and motivation, then jump to the conclusion/discussion for the take-home message. Treat the methods and results sections as optional deep-dives: skim for the main numbers (e.g., a hazard ratio, a percentage), but do not worry about reproducing or fully understanding the statistical machinery. If a paragraph is genuinely unclear after this pass, write down the specific question and bring it to the next supervisor meeting rather than guessing.

Responsibilities

Fellows	Supervisors
Build and organize the data dictionary.	Explain the data source and answer questions about what each variable means.
Ask questions when a variable's meaning is unclear rather than guessing.	Review the data dictionary for accuracy and completeness.
Read the six assigned papers and write the literature review.	Provide the reading list (Table 2.1) and be available for clarifying questions.
Bring specific, written questions about unclear passages to meetings.	Review the literature review for accuracy and give feedback.

Deliverables & Deadline

Due by mid-October 2026:

- Completed data dictionary covering every variable in the project's dataset.
- Short written literature review (one paragraph per paper, plus a synthesis paragraph – roughly 2 pages total).
- Short combined presentation of the data dictionary and literature review to Supervisors.

Estimated Hours

16 hours per Fellow (meetings with the clinical team: 3h; building the data dictionary: 7h; reading and writing the literature review: 4h; presentation prep: 2h).

3 | Module 3: Data Cleaning & Quality Checks

Overview

Real-world data is never perfectly recorded: values go missing, get entered twice, or fall outside a plausible range (a blood pressure of 0, or a typo like 1500 instead of 150). Module 3 introduces *data cleaning*: the process of checking a dataset for these problems and deciding, carefully and transparently, how to handle each one. This is often described as the part of a data project that takes the most time and matters the most – a clean, trustworthy dataset is what makes every later step (including using AI/prediction models in Module 5) meaningful.

Learning Objectives

- Explain the difference between a missing value, an outlier, and a data-entry error, and describe a reasonable way to handle each.
- Use pandas to check a dataset for missing values, duplicate rows, and out-of-range values.
- Understand why a research team documents *every* cleaning decision (what was changed, and why) rather than silently editing data.
- Practice basic descriptive statistics (counts, means, ranges) to sanity-check a dataset before using it.

Tasks

- Complete a guided Python/pandas tutorial on checking for missing values (`isna()`), duplicates (`duplicated()`), and basic summary statistics (`describe()`).
- Using the data dictionary from Module 2, define a plausible range for each numeric variable (e.g., systolic blood pressure between roughly 70 and 250 mmHg) with supervisor guidance, and flag any values that fall outside those ranges.
- Check the dataset for missing values in each column and summarize how much data is missing and where.
- Under supervisor guidance, apply simple, well-documented fixes where appropriate (e.g., standardizing units, correcting an obvious typo) – and leave anything ambiguous flagged for the supervisors to review rather than guessing.

- Write a short data-cleaning log: a plain list of every check performed, what was found, and what (if anything) was changed.

Guiding rule. Never delete or silently change a data point without writing down what you did and why. A good data-cleaning log lets anyone (including a future Fellow) understand exactly how the raw data became the clean data.

Responsibilities

Fellows	Supervisors
Run the missing-value, duplicate, and range checks, and write the cleaning log. Flag ambiguous cases instead of deciding alone.	Confirm plausible ranges for each clinical variable.
Apply only the fixes explicitly approved by supervisors.	Decide how ambiguous or borderline cases should be handled.
	Review the cleaning log for completeness and accuracy.

Deliverables & Deadline

Due by mid-November 2026:

- Python notebook running missing-value, duplicate, and out-of-range checks on the project dataset.
- Data-cleaning log documenting every check and every change made.

Estimated Hours

16 hours per Fellow (pandas tutorial: 4h; running checks: 7h; documenting the cleaning log and review meetings: 5h).

4 | Module 4: Building an Analysis-Ready Dataset & Fall Checkpoint

Overview

With a documented data dictionary (Module 2) and a cleaned dataset (Module 3) in hand, Module 4 has Fellows organize everything into one tidy, well-structured table that the rest of the project can build on – what researchers call an “analysis-ready” dataset. Fellows will also learn, at a simple and practical level, why many health-data research teams organize their data in a standard, consistent way (patient, timestamp, variable, value) rather than an ad hoc spreadsheet layout, so that the same tools and checks can be reused across projects. The module closes with a Fall checkpoint presentation.

Learning Objectives

- Explain what it means for a dataset to be “tidy”: one row per observation, one column per variable, consistent formatting throughout.
- Understand, at a simple level, why standardized data formats (for example, listing each patient’s measurements as *patient ID – date – variable name – value* rather than a different spreadsheet layout for every visit type) make health data easier to check, share, and reuse – the same idea used by real data standards in health research.
- Combine the data dictionary, the cleaned data, and simple derived variables (e.g., pulse pressure = systolic – diastolic) into one organized dataset.
- Communicate progress clearly to a supervisory team that includes both a management researcher and a physician.

Tasks

- Combine the cleaned variables into a single, tidy table with clear column names matching the data dictionary.
- Compute a small number of simple derived variables useful later in the project (e.g., pulse pressure = SBP – DBP; age groups; a flag for whether arterial stiffness was measured at a given visit).
- Reorganize a sample of the dataset into the simple “patient – date – variable – value” long format described above, and discuss with supervisors why this format is useful for health data.

- Write a short data-summary report: how many patients, how many visits, what percentage of visits have each key measurement, and any remaining limitations.
- Prepare and deliver a 10–15 minute Fall checkpoint presentation to Professors Gumus and Daskalopoulou summarizing Modules 1–4: what was learned, what was built, and questions going into the Winter semester.

Responsibilities

Fellows	Supervisors
Build the tidy analysis-ready dataset and the data-summary report.	Confirm which derived variables are useful for the rest of the project.
Prepare and deliver the Fall checkpoint presentation.	Give feedback on the dataset and confirm the Winter-semester direction.
Update the logbook with hours and contributions to date.	Submit the Fall-semester Fellow evaluation to IMSF Program Leadership.

Deliverables & Deadline

Due by mid-December 2026 (end of Fall term):

- Tidy, analysis-ready dataset with derived variables and matching data dictionary.
- Data-summary report.
- Fall checkpoint presentation to both Supervisors.
- Logbook up to date through end of Fall semester (60 hours/Fellow logged).

Estimated Hours

16 hours per Fellow (building the tidy dataset: 8h; summary report: 4h; checkpoint presentation prep: 4h).

5 | Module 5: Applying Existing Prediction Models with Python

Overview

Because arterial stiffness is expensive to measure directly, the research team has developed simple prediction formulas (already derived by the supervisors) that estimate a patient's likely arterial stiffness from routinely available information such as age, sex, and blood pressure. Module 5 does not ask Fellows to derive or invent these formulas – that work is done. Instead, Fellows learn how to *apply* an existing prediction model to real data using Python, and how to check whether its predictions look reasonable. This is one of the most common everyday tasks in applied AI/data-science work: using a model someone else built, correctly and carefully, rather than building one from scratch.

Learning Objectives

- Explain, in plain language, what a prediction formula (or “model”) does: it takes some known inputs (e.g., age, sex, blood pressure) and produces an estimate of something unknown (e.g., arterial stiffness).
- Apply a provided prediction formula to a dataset in Python (a few lines of pandas code, given the formula).
- Compare a model's predictions to reality using simple, intuitive metrics (average error, percentage of predictions within a reasonable range).
- Understand why a prediction model that works well “on average” can still be more or less accurate for different groups of patients.

Tasks

- Review the provided prediction formulas (supplied by supervisors) that estimate arterial stiffness from routinely available patient information.
- Write a short Python function that applies each formula to a row of patient data, and apply it to the whole analysis-ready dataset from Module 4.
- For the subset of patients whose real arterial stiffness *was* measured, compare the model's predictions to the real values, and compute simple accuracy metrics (e.g., average difference, percentage within 1 m/s).
- Build a simple chart (e.g., a scatter plot of predicted vs. actual values) to visualize how well the predictions line up with reality.

- Under supervisor guidance, check whether prediction accuracy differs across simple patient groups (e.g., men vs. women, smokers vs. non-smokers) – and report what you find honestly, including if the model works better for some groups than others.

This is applied AI, not AI research. Fellows are not expected to design, train, or mathematically justify a prediction model. The goal is to build the practical skill of taking an existing formula/model, applying it correctly in code, and evaluating its output critically – exactly what most real-world users of AI tools do.

Responsibilities

Fellows	Supervisors
Implement and apply the prediction formulas in Python.	Provide the prediction formulas and explain what each input means.
Compute accuracy metrics and build the comparison chart.	Confirm which patient subgroups are meaningful to compare.
Report subgroup findings honestly, including weaker results.	Review results and clarify how to interpret them.

Deliverables & Deadline

Due by end of January 2027:

- Python notebook applying the provided prediction formulas to the dataset.
- Accuracy comparison (predicted vs. actual) with chart and simple subgroup breakdown.

Estimated Hours

16 hours per Fellow (reviewing the formulas: 2h; implementation: 7h; accuracy checks and charting: 5h; review meetings: 2h).

6 | Module 6: Using AI Tools to Support Research

Overview

Beyond the project's own prediction formulas, modern AI tools (such as large language model assistants) can help with everyday research tasks: summarizing background literature, organizing notes, drafting plain-language explanations, and helping to spot patterns in text. Module 6 gives Fellows structured, supervised practice using AI tools responsibly as part of a real research project – including understanding their limits, checking their output, and never using them in ways that would violate patient confidentiality.

Learning Objectives

- Understand what AI writing/research assistants are good at (summarizing, organizing, drafting) and where they are unreliable (inventing facts or citations that sound plausible but are wrong).
- Use an AI assistant to help summarize a small set of background research articles, and learn to verify every factual claim against the original source before using it.
- Understand the firm rule that **no real patient data may ever be entered into a public AI tool**, and why this matters for confidentiality.
- Practice using AI tools to support (not replace) the Fellows' own writing and understanding.

Tasks

- Complete a short supervisor-led session on responsible AI use in research: what is appropriate (e.g., summarizing public, already-published articles; drafting an outline for a report) and what is never appropriate (e.g., pasting any patient data, even de-identified, into a public AI tool).
- Use an AI assistant to help produce first-draft plain-language summaries of 3–4 background articles on hypertension or arterial stiffness (provided by supervisors), then fact-check every summary line by line against the original article and correct any errors.
- Use an AI assistant to help organize the Module 1–5 work into a clear, well-structured internal progress document (an outline or table), which Fellows then

review and edit themselves.

- Keep a short log of every instance where an AI tool was used, what it was used for, and what (if anything) had to be corrected – practicing transparency about AI use, a skill increasingly expected in professional and academic work.
- Discuss with supervisors, in a short reflection, one thing the AI tool did well and one thing it got wrong or oversimplified.

Confidentiality rule (non-negotiable). No patient-level data — identified or de-identified — may ever be entered into a public AI/chat tool. AI assistants may only be used on publicly available background material (published articles, guidelines) or on the Fellows’ own already-written notes and drafts.

Responsibilities

Fellows	Supervisors
Fact-check every AI-assisted summary against its source article.	Provide the background articles and lead the responsible-AI-use session.
Keep the AI-use log and write the short reflection.	Confirm the confidentiality rule is understood before any AI tool is used.
Never enter patient data into a public AI tool.	Review the progress document and summaries for accuracy.

Deliverables & Deadline

Due by end of February 2027:

- Fact-checked plain-language summaries of 3–4 background articles.
- AI-use log and short reflection.
- Organized internal progress document (Modules 1–5).

Estimated Hours

14 hours per Fellow (responsible-AI-use session: 2h; drafting and fact-checking summaries: 8h; progress document and reflection: 4h).

7 | Module 7: Data Visualization & Communicating Results

Overview

Good data work is only useful if it can be understood by others. Module 7 focuses on turning the cleaned dataset and prediction-model results from Modules 1–5 into clear charts and a short written summary that supervisors can use as supporting material – for example, an updated figure or table that could help illustrate the project to a broader audience. Fellows practice core data-visualization and communication skills, without being asked to write or edit the research manuscript itself.

Learning Objectives

- Choose an appropriate chart type (bar chart, scatter plot, line chart) for a given comparison.
- Build clear, correctly labeled charts in Python (using matplotlib or seaborn) from the project's cleaned dataset.
- Write a short, plain-language caption for each chart explaining what it shows and why it matters.
- Practice presenting quantitative results to a mixed audience (a management researcher and a physician) clearly and honestly, including any limitations.

Tasks

- Build 3–4 clear, well-labeled charts summarizing the project's data and results so far, for example: the distribution of patient ages and blood pressure; how many patients have each measurement available; and the predicted-vs-actual comparison from Module 5.
- Write a one-sentence, plain-language caption for each chart.
- Assemble the charts and captions into a short (2–3 page) visual summary document, using a consistent, clean style.
- Share the visual summary with supervisors and revise based on their feedback; supervisors will decide whether and how any of this material supports the manuscript that is currently under journal revision.

- Practice explaining one chart out loud, in under two minutes, to a non-technical audience (e.g., a fellow Fellow or a friend) as preparation for Module 8.

Scope note. Fellows produce supporting figures and a plain-language summary only. All decisions about what (if anything) is incorporated into the manuscript under revision are made by the Supervisors.

Responsibilities

Fellows	Supervisors
Build the charts, captions, and visual summary document.	Identify which comparisons are most useful to visualize.
Practice explaining the charts clearly and simply.	Review and approve any material before further use.
Revise charts based on feedback.	Decide whether any material supports the manuscript revision.

Deliverables & Deadline

Due by end of March 2027:

- 3–4 clearly labeled charts with plain-language captions.
- Short (2–3 page) visual summary document.

Estimated Hours

16 hours per Fellow (building charts: 8h; writing captions and the summary document: 5h; practice explaining and review meetings: 3h).

8 | Module 8: Final Presentation & Data Package Handoff

Overview

The final module closes out the Faculty Research Project component of the Fellowship. Fellows consolidate a year of data-collection, data-cleaning, and applied-AI work into a final presentation, hand off a clean and well-documented data package for the research team to continue using, and reflect on how these practical data skills connect to their Impact Initiative and future coursework.

Learning Objectives

- Synthesize eight months of applied data work into a coherent, appropriately-scoped narrative for a general (non-specialist) audience.
- Practice clear, honest communication of both what worked and what did not.
- Reflect on how data collection, data cleaning, and applied use of prediction/AI tools transfer to the Fellows' Impact Initiative and future coursework or careers.
- Complete all administrative closeout steps required by IMSF.

Tasks

- Prepare a final presentation (15–20 minutes) covering: the project's goal, the Fellows' specific contributions (data dictionary, cleaning log, analysis-ready dataset, applied prediction-model comparison, AI-assisted summaries, visual summary), and what was learned.
- Deliver the final presentation to Professors Gumus and Daskalopoulou, and, if scheduled by IMSF, to the broader Fellowship cohort or MDIIM audience.
- Assemble a clean “data package” handoff: the final analysis-ready dataset, the data dictionary, the cleaning log, and all Python notebooks, organized and commented so a future Fellow or research assistant could pick up the work.
- Finalize and submit the completed logbook, documenting all research contributions and hours across both semesters (120 hours/Fellow total).
- Write a short (one-page) reflection connecting the data-skills experience to the Fellows' Impact Initiative and to their broader academic/professional goals.

Responsibilities

Fellows	Supervisors
Prepare and deliver the final presentation.	Provide final feedback and help Fellows frame their contributions accurately.
Assemble and document the final data package handoff.	Confirm the data package is complete and usable going forward.
Submit the completed logbook and reflection.	Submit the end-of-year Fellow performance evaluation to IMSF Program Leadership.

Deliverables & Deadline

Due by end of April 2027 (end of Winter term / Fellowship conclusion):

- Final presentation delivered.
- Complete, documented data package (dataset, data dictionary, cleaning log, notebooks) handed off.
- Completed logbook (120 hours/Fellow, both semesters) and one-page reflection submitted.

Estimated Hours

14 hours per Fellow (presentation prep and delivery: 5h; data package assembly and documentation: 5h; logbook, reflection, and closeout meetings: 4h).

References

- Ben-Shlomo, Y., Spears, M., Boustred, C., May, M., Anderson, S. G., Benjamin, E. J., Boutouyrie, P., Cameron, J., Chen, C.-C., Cruickshank, J. K., et al. (2014). Aortic pulse wave velocity improves cardiovascular event prediction: systematic review and meta-analysis. *Journal of the American College of Cardiology*, 63(7):636–646.
- Butlin, M. and Qasem, A. (2016). Large artery stiffness assessment using SphygmoCor technology. *Pulse*, 4(4):180–192.
- Howard, R. A. (1966). Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1):22–26.
- Protogerou, A., Blacher, J., Stergiou, G. S., Achimastos, A., and Safar, M. E. (2009). Blood pressure response under chronic antihypertensive drug therapy: the role of aortic stiffness in the reason study. *Journal of the American College of Cardiology*, 53(5):445–451.
- Vasan, R. S., Pan, S., Xanthakis, V., Beiser, A., Larson, M. G., Seshadri, S., and Mitchell, G. F. (2022). Arterial stiffness and long-term risk of health outcomes: The framingham heart study. *Hypertension*, 79(5):1045–1056.
- Zargoush, M., Gümüş, M., Verter, V., and Daskalopoulou, S. S. (2018). Designing risk-adjusted therapy for patients with hypertension. *Production and Operations Management*, 27(12):2291–2312.